

# From Discrete Events to Autonomous Agents: The Evolution of Simulation Paradigms in Modeling Complex Systems

## 1. Introduction: The Expanding Role of Simulation

Simulation has evolved from a relatively niche computational technique into one of the most important scientific and engineering methodologies of the modern era. Today, it is widely recognized as a “third pillar of science,” complementing traditional theory and physical experimentation. While theory allows us to derive abstract laws and experimentation helps us test those laws in controlled real-world settings, simulation provides a virtual environment in which complex systems can be constructed, manipulated, and explored with minimal physical risk and cost.

Moreover, researchers and practitioners can use simulation to perform experiments that would otherwise be too expensive, too risky, too time-consuming, or ethically impossible to conduct in real life, for example simulating pandemics, traffic accidents, financial crises, or major industrial failures. The fundamental idea behind simulation is based on the principle of abstraction: identifying the essential elements of a real-world system and representing them in a dynamic computational model. This model can then be executed repeatedly under different assumptions and parameter values to observe how the system behaves over time. While this core principle has remained constant, the paradigms through which simulation is implemented have diversified significantly.

This essay highlights one of the most important developments within simulation methodology: the shift from a process-centric worldview, demonstrated by Discrete Event Simulation (DES), to an agent-centric worldview, embodied by Agent-Based Modeling (ABM). This evolution reflects not only technical changes in modeling approaches, but also a deeper transformation in how complexity is understood and represented. The aim of this essay is to explore this paradigm shift by examining its philosophical foundations, its practical implications for model design, and its real-world applications, with particular emphasis on epidemic modeling during the COVID-19 pandemic. Through this discussion, it becomes clear that the evolution from DES to ABM represents a broader transformation in the way complexity is understood and studied in science and engineering.

## 2. The Process-Centric World: Discrete Event Simulation (DES)

Discrete Event Simulation (DES) is a traditional and historically successful paradigm for simulation in industrial engineering, operations research, logistics, telecommunications, and computer systems. In DES, a system is viewed as a sequence of distinct events that occur at specific points in time. These events change the state of the system instantaneously, while time between events progresses continuously.

Typical DES systems include manufacturing lines, service systems such as banks or call centers, airport operations, supply chains, and computer networks. In a DES model, entities (for example, customers, jobs, or data packets) flow through a network of activities where they may experience delays, undergo processing, and compete for limited resources such as machines, servers, or personnel. The timing of events such as arrivals, departures, machine breakdowns, or service completions governs the system’s evolution. The objective of DES modeling is often to analyze and optimize key performance indicators such as throughput, waiting times, queue lengths, utilization rates, and system efficiency.

*Gerd Wagner’s work on Information and Process Modeling for Simulation (2018; 2019)* provides a systematic framework for understanding DES. He describes the simulated world in terms of four fundamental elements:

- **Objects:** These are the passive or active components of the system, such as machines, servers, vehicles, or buffers. They represent the structural backbone of the system.
- **Events:** These are instantaneous changes in the system state, such as the arrival of a customer, the completion of a service, or the failure of a machine. Events are the primary drivers of change in DES.
- **Activities:** An activity is any tasks that take time and consume resources, for example, manufacturing a part, serving a customer, or transmitting a data packet. Each activity is bounded by start and end events.
- **Processing Networks:** These describe the orchestration of objects, events, and activities into a coherent system. They are often visualized using flowcharts, Business Process Model and Notation (BPMN), or Wagner’s own Discrete Process Modeling Notation (DPMN).

DES is fundamentally global and top-down. The modeller explicitly designs the workflow, including queues, activities, resources, and routing logic. Entities in the system typically follow predefined paths and do not make independent decisions, which makes DES especially effective in systems governed by standard procedures and

process regularity. As a result, DES performs particularly well in environments governed by centralized rules, standardized procedures, and well-defined resource constraints. For example, factory assembly lines, call centers, and computer networks all operate according to fixed rules and well-defined processes. In such settings, DES can provide clear and precise quantitative insights that support practical managerial and engineering decision-making.

### 3. The Agent-Centric World: Agent-Based Modeling (ABM)

In contrast to DES, Agent-Based Modeling (ABM) represents a fundamentally different simulation paradigm. Rather than focusing on overall process flows, ABM takes a bottom-up, individual-centered perspective in which a system is represented as a collection of autonomous, heterogeneous agents. Each agent has attributes, behavioral rules, and the ability to interact with its environment. These agents may represent people, animals, organizations, vehicles, firms, or even abstract decision-making entities. Each agent typically possesses a set of attributes (such as age, health status, wealth, or location), a collection of behavioral rules that govern how it acts, the ability to perceive its local environment, and the capacity to interact with other agents and the environment.

Crucially, no global controller dictates how the system should behave. Instead, system-level behavior emerges naturally from the repeated local interactions among agents. This idea of emergence, where simple individual rules give rise to complex global patterns, is the philosophical cornerstone of ABM.

Several defining characteristics distinguish ABM from DES:

- **Autonomy:** Each agent acts independently without centralized control.
- **Heterogeneity:** Agents can differ in their attributes, preferences, and decision rules.
- **Explicit Space:** Interactions often occur within a physical or networked space, so proximity and connectivity matter.
- **Local Interactions:** Agents make decisions based on local information rather than on global system-wide knowledge.

Because of these properties, ABM is particularly well suited for modeling complex adaptive systems in which adaptation, learning, and self-organization play a central role. Examples include ecological systems, financial markets, urban traffic, social networks, political behavior, and most importantly for this essay, the spread of infectious diseases.

Where DES is typically concerned with optimizing predefined workflows, ABM is primarily concerned with exploring how patterns form, how collective phenomena arise, and how localized interactions scale up to global outcomes. ABM thus provides a computational laboratory for studying systems that are decentralized, adaptive, and behavior-driven.

### 4. The Paradigm Shift: Contrasting Philosophies and Practices

The shift from DES to ABM represents more than a change in modeling technique; it reflects a transition in scientific philosophy and research methodology. *Eldabi et al. (2002)* highlight that simulation modeling can support both quantitative and qualitative research paradigms, and the contrast between DES and ABM illustrates this duality.

DES is deeply rooted in a positivist and quantitative tradition. It seeks to describe systems using mathematically precise structures and to generate numerical performance measures that can be statistically analyzed. The underlying assumption is that the system under study is sufficiently stable, mechanistic, and well-defined that its behavior can be predicted and optimized. This makes DES highly suited for operational research problems, where objectives include improving efficiency, reducing costs, and enhancing throughput.

ABM, while still quantitative in its implementation, aligns more closely with exploratory and theory-building research traditions. Instead of asking only “*What is the optimal configuration of this system?*”, ABM often asks deeper generative questions such as:

*How could this observed global pattern have emerged?*

*Under what conditions does collective behavior change qualitatively?*

*What kinds of individual behaviors are sufficient to reproduce real-world social phenomena?*

Practically, the modeling tasks also differ. In DES, the modeller constructs an accurate process flow by specifying activities, queues, decision points, and resource pools. Once this flow is defined, the model behavior is largely determined. In ABM, the modeller specifies agent rules and interaction structures. The overall system behavior is not programmed directly but instead emerges indirectly from micro-level specifications.

Kevin Dooley (2002) captures this difference by noting that DES is often used for hypothesis testing, for example, determining which queuing policy minimizes average waiting time while ABM is powerful for theory building and conceptual exploration, such as investigating conditions under which social segregation, cooperation, or market bubbles emerge. This philosophical difference explains why ABM has become increasingly important in fields such as sociology, economics, epidemiology, and political science, where human behavior, adaptation, and heterogeneity play dominant roles and cannot be adequately captured by exclusively process-driven models.

## 5. Case in Point: The COVID-19 Pandemic

The COVID-19 pandemic provided a compelling real-world demonstration of the need for agent-based approaches. Early in the pandemic, most governments and research institutions relied on classical compartmental epidemiological models, such as the SIR (Susceptible Infectious Recovered) and SEIR (Susceptible Exposed Infectious Recovered) models. These models divide the population into compartments and describe the flow of individuals between health states using differential equations. Conceptually, this aligns with a process-oriented worldview of DES, where populations “flow” between states according to predefined transition rates.

While these models were vital for rapid initial projections, they suffered from important limitations. Most notably, they assumed homogeneous mixing of the population, meaning every individual was equally likely to interact with every other individual. In reality, however, human interactions are highly structured by age, social networks, geographic location, household composition, workplace organization, and behavioral compliance with public-health policies. These heterogeneities strongly affected how COVID-19 spread.

Agent-based models became particularly useful at this stage. As reported in the *Nature special report by Adam (2020)*, teams such as those at Imperial College London developed detailed ABMs that simulated entire national populations at the level of individuals. Each simulated agent was assigned attributes such as age, household, workplace, school affiliation, and daily movement patterns. Infection then spread through explicit proximity-based interactions between agents in homes, workplaces, schools, and public spaces.

This detail enabled researchers and policymakers to evaluate targeted intervention strategies that were difficult to study with compartmental models alone. For example:

- A DES-style process question might ask:

*“What is the average reduction in transmission resulting from a nationwide lockdown?”*

- An ABM-style agent question might ask:

*“What happens if schools are closed but workplaces remain open? What if only individuals over 65 years of age shelter in place? How do household contacts compare to workplace contacts in driving transmission?”*

Because ABMs represent social networks and daily routines, they can capture the disproportionate impact of COVID-19 on specific demographic groups, the role of super-spreader events, and the complex feedback loop between public behavior and epidemic dynamics. They also revealed the potential for multiple epidemic waves driven by behavioral relaxation and seasonal effects patterns that would later be observed in real data.

In this way, ABM transformed epidemic modeling from an abstract population-level flow problem into a microsimulation of interacting individuals, providing unprecedented insight into the social mechanisms underlying disease spread. The COVID-19 crisis thus marked a moment in which agent-based simulation moved from a mainly academic tool to a central instrument of global public-policy decision-making.

## 6. Synthesis and Convergence: Multi-Paradigm Modeling

The evolution from DES to ABM should not be viewed as replacement, where one paradigm simply supersedes the other. Instead, it is better understood as a process of expansion and integration. Each paradigm remains valuable in its domain of applicability, and modern simulation increasingly relies on multi-paradigm frameworks that combine several approaches within a single model.

Consider, for example, a hospital simulation model:

- **ABM** represents patients and medical staff as autonomous agents who move through the hospital, make decisions, and interact with one another.
- **DES** represents the procedural workflows within operating rooms, emergency departments, and diagnostic laboratories, where standardized processes and resource constraints dominate.
- **Continuous simulation** represents physiological processes such as blood oxygen levels or heart rate in intensive care units.

Such a hybrid model leverages the strengths of each paradigm while compensating for their weaknesses. Wagner's later work on DPMN reflects this unification trend by aiming to integrate object-oriented, process-oriented, and agent-oriented views within a single coherent modeling language.

From a practical standpoint, modern simulation platforms increasingly support this convergence. Tools such as AnyLogic allow DES, ABM, and system dynamics components to coexist and interact. This reflects recognition that real-world systems are rarely followed by a single type of logic. Instead, they consist of interacting subsystems, some best captured by process flows and others by autonomous agents. Therefore, DES and ABM should be understood as complementary tools within a broader simulation toolkit.

## 7. Conclusion

The journey from the process-oriented world of Discrete Event Simulation to the agent-oriented world of Agent-Based Modeling marks a significant maturation in simulation science. It reflects a deeper and more refined understanding of complexity: not all systems can be described as centralized workflows processing passive entities along predefined routes. Many important systems in biology, society, economics, and public health are decentralized, adaptive, and driven by local interactions among autonomous individuals.

DES remains the dominant and indispensable tool for analyzing and optimizing well-defined operational processes, from manufacturing and logistics to computer networks and service systems. Its strengths lie in optimization, efficiency analysis, and quantitative performance evaluation. ABM, on the other hand, has become an essential tool for exploring, understanding, and managing complex adaptive systems, where behavior cannot be centrally prescribed and outcomes emerge from interaction.

The COVID-19 pandemic highlighted the practical importance of this paradigm shift. When confronted with a profoundly complex socio-biological phenomenon, the ability to simulate millions of interacting individuals in a virtual society was not merely an academic exercise; it became a critical instrument of global decision-making. Policymakers relied directly on agent-based simulations to evaluate intervention strategies that shaped the lives of billions of people.

Looking forward, the future of simulation lies not in the dominance of any single paradigm, but in the intelligent integration of multiple paradigms. By combining process-oriented, agent-oriented, and continuous approaches within unified frameworks, simulation will continue to develop as a powerful virtual laboratory for understanding and shaping an increasingly complex world.

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